Question 1: Read dataset and display first 10 rows

```
import pandas as pd

# Load dataset
df = pd.read_csv("bike_details.csv")

# Display first 10 rows
print(df.head(10))

# Shape of dataset
print("Shape:", df.shape)

# Column names
print("Columns:", df.columns)
```

Answer:

This code loads the Bike Details dataset into Pandas DataFrame, displays first 10 rows, dataset shape, and column names.

Question 2: Check missing values

```
# Check missing values
print(df.isnull().sum())
```

Answer:

To handle missing values:

- Numerical columns → fill with mean/median
- Categorical columns → fill with mode

Example:

```
df.fillna(df.median(numeric_only=True), inplace=True)
```

Question 3: Distribution of selling prices

```
import matplotlib.pyplot as plt

plt.hist(df['selling_price'], bins=30)
plt.xlabel("Selling Price")
plt.ylabel("Frequency")
plt.title("Distribution of Selling Price")
plt.show()
```

Answer:

Most bikes are concentrated in lower price ranges, with fewer bikes in higher price ranges.

Question 4: Average selling price by seller type

```
avg_price = df.groupby('seller_type')['selling_price'].mean()

avg_price.plot(kind='bar')
plt.title("Average Selling Price by Seller Type")
plt.show()
```

Observation:

Dealer-listed bikes generally have higher average selling prices compared to individual sellers.

Question 5: Average km_driven by ownership type

```
avg_km = df.groupby('owner')['km_driven'].mean()

avg_km.plot(kind='bar')
plt.title("Average KM Driven by Ownership Type")
plt.show()
```

Answer:

Older ownership categories usually show higher kilometers driven.

Question 6: Remove outliers using IQR

```
Q1 = df['km_driven'].quantile(0.25)
Q3 = df['km_driven'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

print("Before:")
print(df['km_driven'].describe())

# Remove outliers
df_clean = df[(df['km_driven'] >= lower) & (df['km_driven'] <= upper)]

print("After:")
```

```
print(df_clean['km_driven'].describe())
```

Answer:

Outliers were detected using IQR and removed to improve data quality.

Question 7: Scatter plot (Year vs Selling Price)

```
plt.scatter(df['year'], df['selling_price'])
plt.xlabel("Year")
plt.ylabel("Selling Price")
plt.title("Year vs Selling Price")
plt.show()
```

Answer:

Newer bikes generally have higher selling prices than older bikes.

Question 8: One-hot encoding

```
encoded_df = pd.get_dummies(df, columns=['seller_type'])

print(encoded_df.head())
```

Answer:

Categorical seller_type is converted into numerical columns for machine learning.

Question 9: Correlation heatmap

```
import seaborn as sns

corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.title("Correlation Matrix")
plt.show()
```

Answer:

Strong correlations often appear between:

- Year and Selling Price → Positive correlation
- KM Driven and Selling Price → Negative correlation

Question 10: Summary Report

Important factors affecting selling price:

1. Bike age (Year)
2. Kilometers driven
3. Owner type
4. Seller type

Data cleaning performed:

- Missing value treatment
- Outlier removal

Feature engineering performed:

- One-hot encoding of categorical variables